GEE 1104 – CAT 1 Preparation Guide

Key Topics

Module I - Computational Thinking & Specification

Deterministic vs probabilistic systems, Decomposition, Pseudocode, Flowcharts, Specification (Testable, Bounded, Observable), 70/30 Rule.

Module II - AI Foundations & the 2026 AI Stack

History of AI (3 Eras), How LLMS Work, Model Families, 2026 AI Stack, AI in India

Module III - LLM Internals, Prompt Engineering Foundations & Advanced Prompting

Tokens, Context, Temperature, Top-p, System vs User vs Assistant, Reasoning vs Standard Model, Output Shaping, Role, Persona, Few-shot, Chain-of-thought, Reflection, Self Critique, Prompt Chaining.

CAT 1 Preparation Guidelines

1. For preparation of CAT 1, your primary source of reference will be the learning content available in Evolv platform.
2. Additionally, you could follow the steps below, for a quick refresher and to prepare your own notes with help from AI.
3. Let us first pick a topic, say "Deterministic vs Probabilistic Systems".
4. Execute the below prompt in Gemini (recommended https://www.google.com/ai)

   > Under the topic "Computational Thinking & Specification" explain about "Decomposition" in 100 words for a student who does not have any exposure to programming related topic.

5. Read through the response provided and understand the concept.
6. Feel free to ask further questions if needed, but restrict the response to 30, 50 or 100 words, which will provide concise responses that are easily readable.
7. Digest the information, read and form your own understanding.
8. In the exam try to answer the question with your own understanding about the topic in your own words. Marks will not be reduced for grammatical errors.
9. If you need it for future reference, you can copy and paste the response into a central document that will form part of your notes.
10. Repeat the steps 2 to 6 for each Key Topic provided above. Replace "Computational Thinking & Specification" with module heading and replace "Decomposition" with the topic you want to learn.

Sample Questions and Answers

1. Define decomposition in problem-solving, giving one simple example. (Part A – 2 Marks)

Decomposition means breaking a big, complicated problem into smaller, simpler parts that are easier to solve one by one.

Example: Making tea feels like one task, but it is really several small steps — boil water, add tea leaves, add milk and sugar, strain it, pour it into a cup. Solving each small step is much easier than trying to "make tea" all at once in your head.

2. Explain how large language models generate text, describing the role of tokens and probability. (Part B – 10 Marks)

The below diagram shows how LLMs work:

[How LLM works diagram here]

- *On the one side we have the LLM containing billions of parameters. The parameters are actually numbers.*
- *When user gives a prompt, it is converted into tokens, which again becomes numbers.*
- *LLM performs numeric calculation based on the numbers generated from tokens and LLM parameters and a token is predicted, which is what is sent back as generated text.*
- *LLM does not generate the entire response in a single response, it generates the tokens one by one.*

CAT 1 Sample questions for practice

1. Name the three eras in the history of AI, in order. (Part A)
2. Explain what "few-shot prompting" means, with a simple example. (Part B)